Part II, Lecture 5: Loan Pricing, Risk-Adjusted Performance and Pricing Policies

By the end of this lecture you must know how to move from pricing a single loan (Lecture 4) to choosing a bank-wide pricing policy, how the four NewBank policies (base case, risk-adjusted, conservative, expansive) change the interest margin, provisions, portfolio size, RAROC and ROE with and without price elasticity of demand, why full risk-adjusted pricing makes every rating class earn exactly Ke (RAROC = Ke), how portfolio performance is monitored by segment (RAROC, also called ROCA), and how the banker's risk-adjusted pricing framework compares with the corporate WACC.

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1. From One Loan's Price to a Bank-Wide Pricing Policy

Lecture 4 derived the minimum rate on a single loan from the risk-adjusted equilibrium condition: the expected repayment must cover expected loss (PD x LGD), the funding cost on the debt portion (IFTR x (1 - VaR)), the required return on the capital portion (Ke x VaR) and operating costs (C).

Lecture 5 scales the question up: a bank holds a portfolio across rating classes, and must choose a **pricing policy**, that is a rule mapping each class's risk to the rate it is charged. The policy changes not only the margin on each loan but also, once clients react, the **size and risk composition of the whole portfolio**. The performance measures are the RAPMs seen in Lecture 4 (RAROC, EVA), plus the bank-level indicators: interest margin (IM), provisions, operating and net profit, opROA and ROE.

2. The NewBank Exercise: Setup and Policies

The assignment (spreadsheet "Pricing policies implications For Students") gives one bank, **NewBank**, with a current loan portfolio split across **six borrower rating classes**, and all inputs needed: PDs, LGDs, VaR% and exposures per class, the market (non risk-sensitive) interest rate currently applied by all banks including NewBank, NewBank's average borrowing cost, its equity, Ke, tax rate, and fixed operating costs equal for all exposures (per euro of exposure). A **price elasticity of demand (PED) per rating class** is also provided.

NewBank considers four policies, and it is the **only bank changing policy**; all other banks keep the flat, non risk-sensitive market rate:

- **Policy 0, base case:** NewBank keeps the fixed market rate for all classes, disregarding risk differences.
- **Policy 1, risk-adjusted:** NewBank aligns required rates to the risk-adjusted (equilibrium) rates of **all** rating classes.
- **Policy 2, conservative:** NewBank applies risk-adjusted rates only to the **riskier classes 4 to 6**, keeping base-case rates on classes 1 to 3.
- **Policy 3, expansive:** NewBank applies risk-adjusted rates only to the **safer classes 1 to 3** (lower rates), keeping base-case rates on classes 4 to 6.

Each policy is evaluated in two situations: **without PED impact** (clients do not move, volumes fixed) and **with PED impact** (demand per class reacts to the rate charged, so portfolio size and mix change). The model must be built so that setting PED = 0 everywhere reproduces the "without PED" results, and the base case is identical in both panels: these are the spreadsheet's internal checks.

3. Results Without Demand Reaction (PED = 0)

The summary table for a €1,000 million portfolio, all policies at fixed volumes:

Policy	Avg rate	IM (M€)	Provisions (M€)	IM net (M€)	Oper. profit (M€)	Net profit (M€)	opROA	RAROC	ROE
Base case	6.0%	46	23	24	22	17	2%	19%	22%
Risk-adjusted	8.6%	72	24	49	47	36	5%	15%	48%
Conservative	9.2%	78	24	54	52	40	5%	26%	54%

Expansive	5.4%	41	23	18	16	12	2%	8%	17%

Read the mechanism: with captive clients, charging more to risky classes is a pure transfer from borrowers to the bank. Volumes and default rates do not move, so provisions stay flat and higher rates flow almost entirely into the margin. The conservative policy (higher rates only on classes 4-6) posts the highest numbers everywhere. But this world is unrealistic: it assumes risky borrowers accept to keep paying above their risk-adjusted rate while other banks charge them less.

Note the first subtlety: **RAROC falls** under the full risk-adjusted policy (19% to 15%) even as profits rise, because RAROC relates the return to the capital absorbed by risk, and repricing without changing the portfolio leaves the risk structure untouched while revenues change.

4. Results With Price Elasticity of Demand (PED)

Now demand reacts: classes charged more than their equilibrium rate shrink (they leave for the flat-pricing competitors), classes charged less grow.

Policy	Exposure (M€)	Avg rate	IM (M€)	Provisions (M€)	IM net (M€)	Oper. profit (M€)	Net profit (M€)	opROA	RAROC
Base case	1,000	6.0%	46	23	24	22	17	2%	19%
Risk-adjusted	877	6.1%	39	11	28	26	20	2%	15%
Conservative (less loans to risky customers)	647	8.0%	42	11	31	30	23	3%	33%
Expansive (more loans to safer customers)	1,230	5.0%	43	23	20	18	14	2%	9%

The mechanisms, policy by policy:

- **Risk-adjusted:** risky classes 4-6, previously subsidized, now face higher rates and partly leave: exposure falls from 1,000 to 877. The mix improves, so **provisions collapse from 23 to 11**. Net profit rises from 17 to 20 and **ROE from 22% to 27%** despite the smaller balance sheet. This is adverse selection working **for** the bank: underpriced risk exits the portfolio.
- **Conservative:** the risky classes are repriced without offering better rates to the safe ones, so the portfolio shrinks the most, to 647. Provisions fall to 11 and **RAROC reaches 33%**, the best of all policies, on a much smaller bank. ROE 31%.
- **Expansive:** lower rates on safe classes attract volume (exposure 1,230) but margins stay thin: net profit falls to 14 and **ROE to 18%**, below the base case. Growth without profitability.

Key comparison, with vs without PED, risk-adjusted policy: without client reaction the policy looks spectacular (net profit 36, ROE 48%); with realistic demand reaction the same policy delivers a smaller, safer, moderately more profitable bank (net profit 20, ROE 27%). The PED panel, not the fixed-volume panel, is the decision-relevant one.

5. Reading the Results: Volume, Margin, Sustainability

The assignment asks to judge each policy on interest margin, operating profit, portfolio size, RAROC, ROE, and **long-run sustainability**, and to give each policy a nickname.

- **RAROC and ROE diverge:** the conservative policy has the best RAROC (33%) but the risk-adjusted policy keeps a larger franchise (877 vs 647). A policy can maximize return per unit of risk while destroying the bank's size and market presence.
- **Sustainability:** the expansive policy (ROE 18% below base case, thin margins on a bigger book) and the no-PED fantasy of ever-rising margins are not sustainable; the conservative policy's volume collapse raises the same question in the opposite direction. The full risk-adjusted policy is the reference equilibrium.
- **Extension of the exercise:** simulate that the bank wants to keep the **base-case RAROC** while applying the risk-adjusted policy, and comment. This forces you to see that the risk-adjusted rates are pinned down by Ke: changing the target RAROC means changing the pricing rule itself.

6. The Equilibrium Check: RAROC = Ke in Every Class

The spreadsheet states the fundamental consistency check: **in the full risk-adjusted scenario (with and without PED), the RAROC of every rating class must equal Ke**.

This is not a coincidence, it is the definition of the equilibrium rate from Lecture 4 rearranged. RAROC on a loan equals (risk-adjusted return) / (capital absorbed). If the rate iL is set so that expected revenue exactly covers expected loss, funding, operating costs and Ke on the allocated capital, then by construction the return on that capital is exactly Ke. A class whose RAROC exceeds Ke is priced above equilibrium (it will lose clients to competitors); a class below Ke is

priced under its cost of risk (it attracts exactly the clients the bank should not want at that price).

7. The Pricing Incident: Choosing a Policy Against a Flat-Pricing Market

Assignment 5A sets the strategic incident. Our bank has the technology for correct risk-adjusted pricing; the majority of other banks do not, and so far nobody, including us, has practiced it. Four candidate policies are shown as quadrants plotting our required spreads against the flat market average across rating classes, and the table to complete asks for the impact on interest margin, operating profit, portfolio size, RAROC, ROE, long-run sustainability, and a nickname per policy.

The reading grid (developed in the E2 analysis):

- **Pricing above the market where risk is higher (risk-adjusted):** captures fair compensation for risk in sub-investment-grade classes; positive for the interest margin, and it selects better borrowers once clients react.
- **Underpricing high-risk classes relative to risk:** compresses margins and accumulates credit risk; negative.
- **Following the flat market rate:** overpays for low-risk loans and underprices high-risk ones; negative to neutral, and it abandons the technological advantage.

The strategic point mirrors Lecture 4’s adverse selection: the sophisticated bank wins good borrowers with lower rates and lets competitors keep the underpriced bad risks.

8. Portfolio Performance Monitoring: RAROC = ROCA

Assignment 5B is a monitoring exercise on a real portfolio split by segment (Italian banks, other Italian controlled banks, multinationals served in many countries, corporate and investment).

ROCA (return on allocated capital): a synonym of RAROC. Both measure the risk-adjusted return on the capital allocated to a position, a segment or the whole portfolio.

The skill tested: compute and compare segment-level RAROC/ROCA to identify which segments create value (return above Ke on their allocated capital) and which destroy it, exactly as the single-class check of Section 6 but on observed accounting data rather than model prices.

9. The Banker’s Pricing Framework versus WACC

Assignment 5C rewrites the risk-adjusted pricing formula as:

(iL - PD x LGD) x (1 + iL) - C = VaR x Ke + (1 - VaR) x IFTR

and asks to confront it with the corporate finance formula:

WACC = (E/TA) x Ke + (D/TA) x Kd x (1 - T)

Similarities:

- Both are **weighted averages** combining a cost of equity and a cost of debt.
- Both express a **minimum required return**: what must be earned so that all capital providers are remunerated.

Differences, by feature:

Feature	Entity approach (WACC)	Risk-adjusted pricing framework (RAPF)
Primary metric	Discount rate for firm value	Minimum loan rate per transaction
Financial structure	Book weights E/TA and D/TA	Economic capital weights VaR and 1 - VaR
Time horizon	Long-run, whole firm	Single transaction (one-year loan)
Valuation basis	Enterprise valuation	Per-loan equilibrium, net of expected loss PD x LGD and costs C
Tax treatment	After-tax cost of debt Kd x (1 - T)	IFTR set by treasury, no tax shield inside the formula
Key use	Investment and valuation decisions	Loan pricing and accept/reject decisions

The deep link: VaR (the capital allocated to the loan) plays the role of E/TA (the equity share of the firm). The banker’s formula is a WACC computed at the level of a single loan, with the expected loss stripped out of the repayment on the left-hand side.

10. Common Mistakes and Exam Traps

- Reading the without-PED panel as the policy’s real outcome: it assumes clients never react. Decision-relevant results are always the **with-PED** numbers.
- Confusing the conservative and expansive policies: conservative re-prices **up only the riskier classes 4-6** (less

loans to risky customers), expansive re-prices **down only the safer classes 1-3** (more loans to safer customers).

- Believing the policy with the highest margin is the best: without PED the conservative policy maximizes everything, but with PED it shrinks the portfolio to 647. Margin, volume and sustainability must be read together.
- Mixing up RAROC and ROE rankings: with PED the conservative policy has the best RAROC (33%) but the risk-adjusted policy preserves a much larger bank; the expansive policy grows the portfolio yet destroys ROE (18% < 22% base case).
- Forgetting the equilibrium check: under full risk-adjusted pricing, **every class's RAROC must equal Ke**; a class above or below signals a pricing error, not an opportunity.
- Forgetting the model's internal checks: PED = 0 everywhere must reproduce the without-PED results, and the base case must be identical in both panels.
- Treating ROCA and RAROC as two different indicators: they are synonyms (return on allocated capital).
- Mapping the RAPF onto WACC incorrectly: the equity weight in the banker's formula is **VaR** (allocated capital on the single loan), not the bank's book equity ratio, and the debt leg uses the gross IFTR, not an after-tax Kd.

11. Must Know Absolutely

Must know:
• The four NewBank policies: base case (flat market rate), risk-adjusted (equilibrium rates on all classes), conservative (risk-adjusted only on risky classes 4-6), expansive (risk-adjusted only on safe classes 1-3).
• With PED: risk-adjusted shrinks the portfolio to 877, halves provisions (23 to 11), raises ROE to 27%; conservative shrinks it to 647 with the best RAROC (33%); expansive grows it to 1,230 but cuts ROE to 18% below the base case.
• Without PED the numbers are inflated and misleading: repricing with captive clients is a pure transfer, not a real outcome.
• Equilibrium check: under full risk-adjusted pricing, RAROC of every rating class equals Ke, by construction of the pricing formula.
• ROCA = RAROC (return on allocated capital), used for segment-level portfolio monitoring.
• RAPF vs WACC: same weighted-average logic (cost of equity + cost of debt), but VaR weights instead of book E/TA, gross IFTR instead of after-tax Kd, single-transaction horizon, and explicit deduction of expected loss PD x LGD and operating costs C.
• Strategic reading (5A): a bank with risk-adjusted pricing technology wins good borrowers with lower rates and leaves underpriced bad risks to flat-pricing competitors.